

CSS

Transition, transform, animation

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Transform</title>
    <style>
      section {
        border: 2px solid black;
        /* height: 80vh; */
        /* position: relative; */
      }
      div {
        width: 200px;
        height: 200px;
        background-color: gray;
        /* transition-property: background-color, color, transform; */
        /* transition-property: all; */
        /* transition-duration: 2s; */
        /* transition-delay: 2s; */
        /* transition-timing-function: ease-in ; */
        transition: all 2s ease-in 2s;

        /* transform: translateX(-100px); move elements */
        /* transform: translateY(100px); */
        /* position: absolute;
          top: 50%;
          left: 50%; */
        /* transform: translate(-50%, -50%); */

        /* transform: rotateX(45deg); */
        /* transform: rotateY(45deg); */
        /* transform: rotate(45deg); */
        /* transform: scaleX(1.5); */
        /* transform: scaleY(1.5); */
        /* transform: scale(1.5); */
        /* transform: skewX(50deg); */
        /* transform: skewY(50deg); */
        /* transform: skew(50deg); */
      }
      div:hover {
        transform: scale(1.2) rotate(45deg);
        background-color: aqua;
        color: white;
      }
    </style>
  </head>
  <body>
    <div></div>
  </body>
</html>
```

```

}

/* transform => allows to move, scale, or rotate elements */
header {
  /* animation-name: rotateElement;
  animation-duration: 5s;
  animation-iteration-count: infinite; */
  display: inline-block;
  /* animation-timing-function: ease-in;
  animation-direction: alternate-reverse; */
  animation: rotateElement 5s ease-in 2s infinite ;
}
header:hover{
  animation-play-state: paused;
}
@keyframes rotateElement {
  /* 0% {
    background-color: green;
  }
  25%{
    background-color: blue;
  }
  50%{
    background-color: red;
  }
  75%{
    background-color: orange;
    transform: translateX(100px);
  }
  100% {
    background-color: yellow;
  } */

  /* from{
    transform: translateX(0);
  }
  to{
    transform: translateX(200px);
  } */
  0% {
    transform: translateY(0);
  }
  /* 50%{
    transform: translateY(-50px);
  } */
  100%{
    transform: translateY(200px);
  }
}

```

```

    }
  </style>
</head>
<body>
  <header>
    <h1>Animation</h1>
  </header>

  <h1>Transform</h1>

  <section>
    <div>Hello there</div>
  </section>
</body>
</html>

```

Theory Notes

1. Transform

- **Definition:** CSS transform allows you to move, rotate, scale, and skew elements.
- **Common functions:**
 - `translate(x, y)` → moves element.
 - `rotate(deg)` → rotates element.
 - `scale(x, y)` → resizes element.
 - `skew(x, y)` → slants element.
- Example:

```

div:hover {

  transform: scale(1.2) rotate(45deg);

}

```

2. Transition

- **Definition:** Smoothly animates property changes.
- **Properties:**
 - `transition-property` → which property to animate.
 - `transition-duration` → how long animation lasts.
 - `transition-delay` → when animation starts.
 - `transition-timing-function` → speed curve (ease, linear, ease-in, ease-out).
- Example:

```

div {

  transition: all 2s ease-in 2s;

}

```

}

3. Animation

- **Definition:** Keyframe-based animations for complex effects.
- **Properties:**
 - animation-name → name of keyframes.
 - animation-duration → how long animation runs.
 - animation-iteration-count → number of repeats (infinite).
 - animation-direction → normal, reverse, alternate.
 - animation-play-state → running or paused.
- **Keyframes:**

```
@keyframes rotateElement {  
  
  0% { transform: translateY(0); }  
  
  100% { transform: translateY(200px); }  
  
}
```

4. Differences

- **Transform** → changes element's shape/position instantly.
- **Transition** → smooth change between states (e.g., hover).
- **Animation** → continuous or complex movement defined by keyframes.

Practice Tasks

Short Answer

1. What is the difference between transition and animation?
2. How does ease-in differ from linear in transitions?
3. What does animation-play-state: paused do?

Coding Exercises

1. Create a box that scales up and changes color on hover using transition.
2. Animate a header that moves left to right continuously.
3. Build a button that rotates 360° when hovered.
4. Use keyframes to animate a bouncing ball effect.
5. Create an animation that alternates between two colors.
6. Implement a pause-on-hover effect for a running animation.
7. Combine scale and rotate transforms in a single animation.